18 Review Problems

1. **Probability Rules and Axioms.** Make a list of all the probability rules and axioms we have learned. List them by name and using good probability notation.

2. **Mutually exclusive vs independent.**

   a. What does it mean for events A and B to be **mutually exclusive**? Express this in words an in probility notation.
   b. What does it mean for events A and B to be **independent**? Express this in words an in probility notation.
   c. Is it possible for events A and B to be mutually exclusive and independent? Give an example where this happens or explain why it cannot happen.
   d. Give an example of a probability rule that only holds if events are mutually exclusive.
   e. Give an example of a probability rule that only holds if events are independent.

3. a. What is the difference between $p(A \mid B)$ and $p(B \mid A)$?
   b. Give an example where these are $p(A \mid B) = p(B \mid A)$.
   c. Give a different example where they $p(A \mid B) \neq p(B \mid A)$.
   d. True or False: If A and B are independent, then $p(A \mid B) = p(B \mid A)$? (Either explain why this is always true or give an example where it is false.)

4. **Binomial distributions**

   a. For random variable to be a binomial random variable, what 5 things must be true?
   b. Give an example of a situation that is $\mathsf{Binom}(20, 0.25)$.

5. Suppose $X \sim \mathsf{Binom}(20, 0.25)$. (a) What is $p(X = 4)$? (b) What is $\mathrm{E}(X)$?

6. Suppose at a certain college 60% of students are female, 10% of female students are from international students, and 12% of male students are international. What percentage of the international students are female?

   a. Do you expect the answer to be less than 12%, between 12% and 60%, or over 60%? Explain.
   b. Express the three numbers in this problem and the number asked for in the question using good probability notation.
   c. Answer the answer numerically.

7. Suppose that we flip a fair coin until either it comes up tails twice or we have flipped it six times.

   a. Is this a binomial situation? Explain.
   b. What is the expected number of times we flip the coin?

8. Suppose that we flip a fair coin until it comes up tails twice, no matter how many tries that takes.

   a. Is this a binomial situation? Explain.
   b. What is the expected number of times we flip the coin?

9. A bowl of M&M's is almost empty. There are 3 reds, 5 blues, and 8 browns. You randomly select three.

   a. What is the probability that you get one of each color?
   b. What is the probability that they are all brown?
   c. What is the probability that they are all red?
   d. What is the probability that they are all the same color?
   e. If they are all the same color, what is the probability that they are all brown?
   f. What is the probability that you get 2 browns and one blue?
   g. What is the probability that you get two of one color and one of a different color?
   h. What is the expected number of brown candies you select?